parameter to use different AI models while maintaining consistent functionality across the system. This flexibility allows for rapid experimentation and optimization without significant code changes.

To explore this functionality hands-on, I've included the complete example in the source code accompanying this chapter. You can find the project in the ch04/chat-rsc-tool-calls folder. To run the example, follow these steps:

1. Navigate to the root folder of the project repository.
2. Execute the following command:

```
$ npm run dev -w ch04/chat-rsc-tool-calls
```

Open your web browser and visit the local development server address at `http://localhost:3000`.

Once on the chat page, you can interact with the LLM by asking about the current weather in any city worldwide. The system will render a weather report based on real-time data. Feel free to inquire about the weather in your local city!

You are now equipped with the tools and techniques to utilize the Vercel AI SDK to build interactive generative AI web apps with ease. In the next chapter, we take it to another level by introducing another great tool that integrates well with our framework: *LangChain*.

Summary

- RSCs enable server-side rendering of specific components, improving performance and simplifying data fetching in AI-powered applications.
- The Vercel AI SDK provides helpers like `createAI`, `getMutableAIState`, `useActions`, `useUIState`, and `useAIState` for managing AI and UI state in React applications.
- Separating AI state and UI state allows for more efficient handling of data, improved performance, and better synchronization between server and client.
- Structured data generation techniques like prompt engineering, function calling, and Zod schema validation enable AI models to produce organized, type-safe data instead of raw text.
- The `generateObject` and `streamObject` functions in the Vercel AI SDK facilitate the creation of type-safe, structured data from AI responses, enhancing reliability and usability.
- Tool-calling and function-calling capabilities allow AI models to interact with external resources, expanding their abilities beyond their initial training data.
- Implementing custom tools with the Vercel AI SDK involves defining functions, registering them with the SDK, and handling their responses as React components.
- The Vercel AI SDK's common interface for AI model interactions enables easy swapping of different AI models while maintaining consistent functionality across the system.